

ONNX Model Report

- **Model Path:** C:\Users\Administrator\Desktop\hdkx-aicompile-1\resnet18.onnx
- **IR Version:** 6
- **Opset Import:** ['11']
- **Producer:** pytorch (2.5.1)

Inputs

Name	Shape	Type
input	[batch_size, 3, 224, 224]	FLOAT

Outputs

Name	Shape	Type
output	[batch_size, 1000]	FLOAT

Initializers (Weights)

Total Initializers: 42

Name	Shape	Type	Stats
fc.weight	[1000, 512]	FLOAT	min:-0.28, max:0.72
fc.bias	[1000]	FLOAT	min:-0.05, max:0.06
onnx::Conv_193	[64, 3, 7, 7]	FLOAT	min:-0.31, max:0.39
onnx::Conv_194	[64]	FLOAT	min:-0.64, max:0.69
onnx::Conv_196	[64, 64, 3, 3]	FLOAT	min:-0.37, max:0.33
onnx::Conv_197	[64]	FLOAT	min:-0.82, max:1.10

Name	Shape	Type	Stats
onnx::Conv_230	[256]	FLOAT	min:-0.34, max:0.24
onnx::Conv_232	[256, 256, 3, 3]	FLOAT	min:-0.22, max:0.27
onnx::Conv_233	[256]	FLOAT	min:-0.81, max:0.74
onnx::Conv_235	[256, 256, 3, 3]	FLOAT	min:-0.97, max:0.73
onnx::Conv_236	[256]	FLOAT	min:-1.01, max:1.06
onnx::Conv_238	[512, 256, 3, 3]	FLOAT	min:-0.15, max:0.30
onnx::Conv_239	[512]	FLOAT	min:-0.48, max:0.56
onnx::Conv_241	[512, 512, 3, 3]	FLOAT	min:-0.69, max:1.14
onnx::Conv_242	[512]	FLOAT	min:-1.13, max:0.75
onnx::Conv_244	[512, 256, 1, 1]	FLOAT	min:-0.83, max:1.00
onnx::Conv_245	[512]	FLOAT	min:-0.76, max:0.26
onnx::Conv_247	[512, 512, 3, 3]	FLOAT	min:-0.18, max:0.29
onnx::Conv_248	[512]	FLOAT	min:-0.85, max:0.51
onnx::Conv_250	[512, 512, 3, 3]	FLOAT	min:-2.33, max:3.65
onnx::Conv_251	[512]	FLOAT	min:-0.70, max:2.23

Computation Graph (Nodes)

Total Nodes: 49

Node 0: Conv

- **Name:** /conv1/Conv
- **Inputs:**
 - input

- onnx::Conv_193

- onnx::Conv_194

- **Outputs:**

- /conv1/Conv_output_0

- **Attributes:**

- dilations : [1, 1]

- group : 1

- kernel_shape : [7, 7]

- pads : [3, 3, 3, 3]

- strides : [2, 2]

Node 1: Relu

- **Name:** /relu/Relu

- **Inputs:**

- /conv1/Conv_output_0

- **Outputs:**

- /relu/Relu_output_0

Node 2: MaxPool

- **Name:** /maxpool/MaxPool

- **Inputs:**

- /relu/Relu_output_0

- **Outputs:**

- /maxpool/MaxPool_output_0

- **Attributes:**

- ceil_mode : 0

- dilations : [1, 1]

- kernel_shape : [3, 3]

- pads : [1, 1, 1, 1]

- strides : [2, 2]

Node 3: Conv

- **Name:** /layer1/layer1.0/conv1/Conv
- **Inputs:**
 - /maxpool/MaxPool_output_0
 - onnx::Conv_196
 - onnx::Conv_197
- **Outputs:**
 - /layer1/layer1.0/conv1/Conv_output_0
- **Attributes:**
 - dilations : [1, 1]
 - group : 1
 - kernel_shape : [3, 3]
 - pads : [1, 1, 1, 1]
 - strides : [1, 1]

Node 4: Relu

- **Name:** /layer1/layer1.0/relu/Relu
- **Inputs:**
 - /layer1/layer1.0/conv1/Conv_output_0
- **Outputs:**
 - /layer1/layer1.0/relu/Relu_output_0

Node 5: Conv

- **Name:** /layer1/layer1.0/conv2/Conv
- **Inputs:**
 - /layer1/layer1.0/relu/Relu_output_0
 - onnx::Conv_199
 - onnx::Conv_200
- **Outputs:**
 - /layer1/layer1.0/conv2/Conv_output_0

- **Attributes:**

- `dilations` : [1, 1]
- `group` : 1
- `kernel_shape` : [3, 3]
- `pads` : [1, 1, 1, 1]
- `strides` : [1, 1]

Node 6: `Add`

- **Name:** /layer1/layer1.0/Add

- **Inputs:**

- /layer1/layer1.0/conv2/Conv_output_0
- /maxpool/MaxPool_output_0

- **Outputs:**

- /layer1/layer1.0/Add_output_0

Node 7: `Relu`

- **Name:** /layer1/layer1.0/relu_1/Relu

- **Inputs:**

- /layer1/layer1.0/Add_output_0

- **Outputs:**

- /layer1/layer1.0/relu_1/Relu_output_0

Node 8: `Conv`

- **Name:** /layer1/layer1.1/conv1/Conv

- **Inputs:**

- /layer1/layer1.0/relu_1/Relu_output_0
- `onnx::Conv_202`
- `onnx::Conv_203`

- **Outputs:**

- /layer1/layer1.1/conv1/Conv_output_0

- **Attributes:**

- `dilations` : [1, 1]
- `group` : 1
- `kernel_shape` : [3, 3]
- `pads` : [1, 1, 1, 1]
- `strides` : [1, 1]

Node 9: Relu

- **Name:** /layer1/layer1.1/relu/Relu

- **Inputs:**

- /layer1/layer1.1/conv1/Conv_output_0

- **Outputs:**

- /layer1/layer1.1/relu/Relu_output_0

Node 10: Conv

- **Name:** /layer1/layer1.1/conv2/Conv

- **Inputs:**

- /layer1/layer1.1/relu/Relu_output_0
- `onnx::Conv_205`
- `onnx::Conv_206`

- **Outputs:**

- /layer1/layer1.1/conv2/Conv_output_0

- **Attributes:**

- `dilations` : [1, 1]
- `group` : 1
- `kernel_shape` : [3, 3]
- `pads` : [1, 1, 1, 1]
- `strides` : [1, 1]

Node 11: Add

- **Name:** /layer1/layer1.1/Add

- **Inputs:**

- /layer1/layer1.1/conv2/Conv_output_0
- /layer1/layer1.0/relu_1/Relu_output_0

- **Outputs:**

- /layer1/layer1.1/Add_output_0

Node 12: Relu

- **Name:** /layer1/layer1.1/relu_1/Relu

- **Inputs:**

- /layer1/layer1.1/Add_output_0

- **Outputs:**

- /layer1/layer1.1/relu_1/Relu_output_0

Node 13: Conv

- **Name:** /layer2/layer2.0/conv1/Conv

- **Inputs:**

- /layer1/layer1.1/relu_1/Relu_output_0
- onnx::Conv_208
- onnx::Conv_209

- **Outputs:**

- /layer2/layer2.0/conv1/Conv_output_0

- **Attributes:**

- dilations : [1, 1]
- group : 1
- kernel_shape : [3, 3]
- pads : [1, 1, 1, 1]
- strides : [2, 2]

Node 14: Relu

- **Name:** /layer2/layer2.0/relu/Relu

- **Inputs:**

- /layer2/layer2.0/conv1/Conv_output_0

- **Outputs:**

- /layer2/layer2.0/relu/Relu_output_0

Node 15: Conv

- **Name:** /layer2/layer2.0/conv2/Conv

- **Inputs:**

- /layer2/layer2.0/relu/Relu_output_0

- onnx::Conv_211

- onnx::Conv_212

- **Outputs:**

- /layer2/layer2.0/conv2/Conv_output_0

- **Attributes:**

- dilations : [1, 1]

- group : 1

- kernel_shape : [3, 3]

- pads : [1, 1, 1, 1]

- strides : [1, 1]

Node 16: Conv

- **Name:** /layer2/layer2.0/downsample/downsample.0/Conv

- **Inputs:**

- /layer1/layer1.1/relu_1/Relu_output_0

- onnx::Conv_214

- onnx::Conv_215

- **Outputs:**

- /layer2/layer2.0/downsample/downsample.0/Conv_output_0

- **Attributes:**

- dilations : [1, 1]

- `group` : 1
- `kernel_shape` : [1, 1]
- `pads` : [0, 0, 0, 0]
- `strides` : [2, 2]

Node 17: Add

- **Name:** /layer2/layer2.0/Add
- **Inputs:**
 - /layer2/layer2.0/conv2/Conv_output_0
 - /layer2/layer2.0/downsample/downsample.0/Conv_output_0
- **Outputs:**
 - /layer2/layer2.0/Add_output_0

Node 18: Relu

- **Name:** /layer2/layer2.0/relu_1/Relu
- **Inputs:**
 - /layer2/layer2.0/Add_output_0
- **Outputs:**
 - /layer2/layer2.0/relu_1/Relu_output_0

Node 19: Conv

- **Name:** /layer2/layer2.1/conv1/Conv
- **Inputs:**
 - /layer2/layer2.0/relu_1/Relu_output_0
 - onnx::Conv_217
 - onnx::Conv_218
- **Outputs:**
 - /layer2/layer2.1/conv1/Conv_output_0
- **Attributes:**
 - `dilations` : [1, 1]

- `group` : 1
- `kernel_shape` : [3, 3]
- `pads` : [1, 1, 1, 1]
- `strides` : [1, 1]

Node 20: Relu

- **Name:** /layer2/layer2.1/relu/Relu
- **Inputs:**
 - /layer2/layer2.1/conv1/Conv_output_0
- **Outputs:**
 - /layer2/layer2.1/relu/Relu_output_0

Node 21: Conv

- **Name:** /layer2/layer2.1/conv2/Conv
- **Inputs:**
 - /layer2/layer2.1/relu/Relu_output_0
 - onnx::Conv_220
 - onnx::Conv_221
- **Outputs:**
 - /layer2/layer2.1/conv2/Conv_output_0
- **Attributes:**
 - `dilations` : [1, 1]
 - `group` : 1
 - `kernel_shape` : [3, 3]
 - `pads` : [1, 1, 1, 1]
 - `strides` : [1, 1]

Node 22: Add

- **Name:** /layer2/layer2.1/Add
- **Inputs:**
 - /layer2/layer2.1/conv2/Conv_output_0

- /layer2/layer2.0/relu_1/Relu_output_0

- **Outputs:**

- /layer2/layer2.1/Add_output_0

Node 23: Relu

- **Name:** /layer2/layer2.1/relu_1/Relu

- **Inputs:**

- /layer2/layer2.1/Add_output_0

- **Outputs:**

- /layer2/layer2.1/relu_1/Relu_output_0

Node 24: Conv

- **Name:** /layer3/layer3.0/conv1/Conv

- **Inputs:**

- /layer2/layer2.1/relu_1/Relu_output_0
- onnx::Conv_223
- onnx::Conv_224

- **Outputs:**

- /layer3/layer3.0/conv1/Conv_output_0

- **Attributes:**

- dilations : [1, 1]
- group : 1
- kernel_shape : [3, 3]
- pads : [1, 1, 1, 1]
- strides : [2, 2]

Node 25: Relu

- **Name:** /layer3/layer3.0/relu/Relu

- **Inputs:**

- /layer3/layer3.0/conv1/Conv_output_0

- **Outputs:**

- /layer3/layer3.0/relu/Relu_output_0

Node 26: Conv

- **Name:** /layer3/layer3.0/conv2/Conv

- **Inputs:**

- /layer3/layer3.0/relu/Relu_output_0
- onnx::Conv_226
- onnx::Conv_227

- **Outputs:**

- /layer3/layer3.0/conv2/Conv_output_0

- **Attributes:**

- dilations : [1, 1]
- group : 1
- kernel_shape : [3, 3]
- pads : [1, 1, 1, 1]
- strides : [1, 1]

Node 27: Conv

- **Name:** /layer3/layer3.0/downsample/downsample.0/Conv

- **Inputs:**

- /layer2/layer2.1/relu_1/Relu_output_0
- onnx::Conv_229
- onnx::Conv_230

- **Outputs:**

- /layer3/layer3.0/downsample/downsample.0/Conv_output_0

- **Attributes:**

- dilations : [1, 1]
- group : 1
- kernel_shape : [1, 1]
- pads : [0, 0, 0, 0]

- `strides` : [2, 2]

Node 28: Add

- **Name:** /layer3/layer3.0/Add
- **Inputs:**
 - /layer3/layer3.0/conv2/Conv_output_0
 - /layer3/layer3.0/downsample/downsample.0/Conv_output_0
- **Outputs:**
 - /layer3/layer3.0/Add_output_0

Node 29: Relu

- **Name:** /layer3/layer3.0/relu_1/Relu
- **Inputs:**
 - /layer3/layer3.0/Add_output_0
- **Outputs:**
 - /layer3/layer3.0/relu_1/Relu_output_0

Node 30: Conv

- **Name:** /layer3/layer3.1/conv1/Conv
- **Inputs:**
 - /layer3/layer3.0/relu_1/Relu_output_0
 - `onnx::Conv_232`
 - `onnx::Conv_233`
- **Outputs:**
 - /layer3/layer3.1/conv1/Conv_output_0
- **Attributes:**
 - `dilations` : [1, 1]
 - `group` : 1
 - `kernel_shape` : [3, 3]
 - `pads` : [1, 1, 1, 1]

- `strides` : [1, 1]

Node 31: Relu

- **Name:** /layer3/layer3.1/relu/Relu
- **Inputs:**
 - /layer3/layer3.1/conv1/Conv_output_0
- **Outputs:**
 - /layer3/layer3.1/relu/Relu_output_0

Node 32: Conv

- **Name:** /layer3/layer3.1/conv2/Conv
- **Inputs:**
 - /layer3/layer3.1/relu/Relu_output_0
 - onnx::Conv_235
 - onnx::Conv_236
- **Outputs:**
 - /layer3/layer3.1/conv2/Conv_output_0
- **Attributes:**
 - `dilations` : [1, 1]
 - `group` : 1
 - `kernel_shape` : [3, 3]
 - `pads` : [1, 1, 1, 1]
 - `strides` : [1, 1]

Node 33: Add

- **Name:** /layer3/layer3.1/Add
- **Inputs:**
 - /layer3/layer3.1/conv2/Conv_output_0
 - /layer3/layer3.0/relu_1/Relu_output_0
- **Outputs:**

- /layer3/layer3.1/Add_output_0

Node 34: Relu

- **Name:** /layer3/layer3.1/relu_1/Relu
- **Inputs:**
 - /layer3/layer3.1/Add_output_0
- **Outputs:**
 - /layer3/layer3.1/relu_1/Relu_output_0

Node 35: Conv

- **Name:** /layer4/layer4.0/conv1/Conv
- **Inputs:**
 - /layer3/layer3.1/relu_1/Relu_output_0
 - onnx::Conv_238
 - onnx::Conv_239
- **Outputs:**
 - /layer4/layer4.0/conv1/Conv_output_0
- **Attributes:**
 - dilations : [1, 1]
 - group : 1
 - kernel_shape : [3, 3]
 - pads : [1, 1, 1, 1]
 - strides : [2, 2]

Node 36: Relu

- **Name:** /layer4/layer4.0/relu/Relu
- **Inputs:**
 - /layer4/layer4.0/conv1/Conv_output_0
- **Outputs:**
 - /layer4/layer4.0/relu/Relu_output_0

Node 37: Conv

- **Name:** /layer4/layer4.0/conv2/Conv
- **Inputs:**
 - /layer4/layer4.0/relu/Relu_output_0
 - onnx::Conv_241
 - onnx::Conv_242
- **Outputs:**
 - /layer4/layer4.0/conv2/Conv_output_0
- **Attributes:**
 - dilations : [1, 1]
 - group : 1
 - kernel_shape : [3, 3]
 - pads : [1, 1, 1, 1]
 - strides : [1, 1]

Node 38: Conv

- **Name:** /layer4/layer4.0/downsample/downsample.0/Conv
- **Inputs:**
 - /layer3/layer3.1/relu_1/Relu_output_0
 - onnx::Conv_244
 - onnx::Conv_245
- **Outputs:**
 - /layer4/layer4.0/downsample/downsample.0/Conv_output_0
- **Attributes:**
 - dilations : [1, 1]
 - group : 1
 - kernel_shape : [1, 1]
 - pads : [0, 0, 0, 0]
 - strides : [2, 2]

Node 39: Add

- **Name:** /layer4/layer4.0/Add
- **Inputs:**
 - /layer4/layer4.0/conv2/Conv_output_0
 - /layer4/layer4.0/downsample/downsample.0/Conv_output_0
- **Outputs:**
 - /layer4/layer4.0/Add_output_0

Node 40: Relu

- **Name:** /layer4/layer4.0/relu_1/Relu
- **Inputs:**
 - /layer4/layer4.0/Add_output_0
- **Outputs:**
 - /layer4/layer4.0/relu_1/Relu_output_0

Node 41: Conv

- **Name:** /layer4/layer4.1/conv1/Conv
- **Inputs:**
 - /layer4/layer4.0/relu_1/Relu_output_0
 - onnx::Conv_247
 - onnx::Conv_248
- **Outputs:**
 - /layer4/layer4.1/conv1/Conv_output_0
- **Attributes:**
 - dilations : [1, 1]
 - group : 1
 - kernel_shape : [3, 3]
 - pads : [1, 1, 1, 1]
 - strides : [1, 1]

Node 42: Relu

- **Name:** /layer4/layer4.1/relu/Relu
- **Inputs:**
 - /layer4/layer4.1/conv1/Conv_output_0
- **Outputs:**
 - /layer4/layer4.1/relu/Relu_output_0

Node 43: Conv

- **Name:** /layer4/layer4.1/conv2/Conv
- **Inputs:**
 - /layer4/layer4.1/relu/Relu_output_0
 - onnx::Conv_250
 - onnx::Conv_251
- **Outputs:**
 - /layer4/layer4.1/conv2/Conv_output_0
- **Attributes:**
 - dilations : [1, 1]
 - group : 1
 - kernel_shape : [3, 3]
 - pads : [1, 1, 1, 1]
 - strides : [1, 1]

Node 44: Add

- **Name:** /layer4/layer4.1/Add
- **Inputs:**
 - /layer4/layer4.1/conv2/Conv_output_0
 - /layer4/layer4.0/relu_1/Relu_output_0
- **Outputs:**
 - /layer4/layer4.1/Add_output_0

Node 45: Relu

- **Name:** /layer4/layer4.1/relu_1/Relu
- **Inputs:**
 - /layer4/layer4.1/Add_output_0
- **Outputs:**
 - /layer4/layer4.1/relu_1/Relu_output_0

Node 46: GlobalAveragePool

- **Name:** /avgpool/GlobalAveragePool
- **Inputs:**
 - /layer4/layer4.1/relu_1/Relu_output_0
- **Outputs:**
 - /avgpool/GlobalAveragePool_output_0

Node 47: Flatten

- **Name:** /Flatten
- **Inputs:**
 - /avgpool/GlobalAveragePool_output_0
- **Outputs:**
 - /Flatten_output_0
- **Attributes:**
 - axis : 1

Node 48: Gemm

- **Name:** /fc/Gemm
- **Inputs:**
 - /Flatten_output_0
 - fc.weight
 - fc.bias

- **Outputs:**

- output

- **Attributes:**

- alpha : 1.0
- beta : 1.0
- transB : 1